

Laravel Relationships

Diagram se samjhao - database, model, output

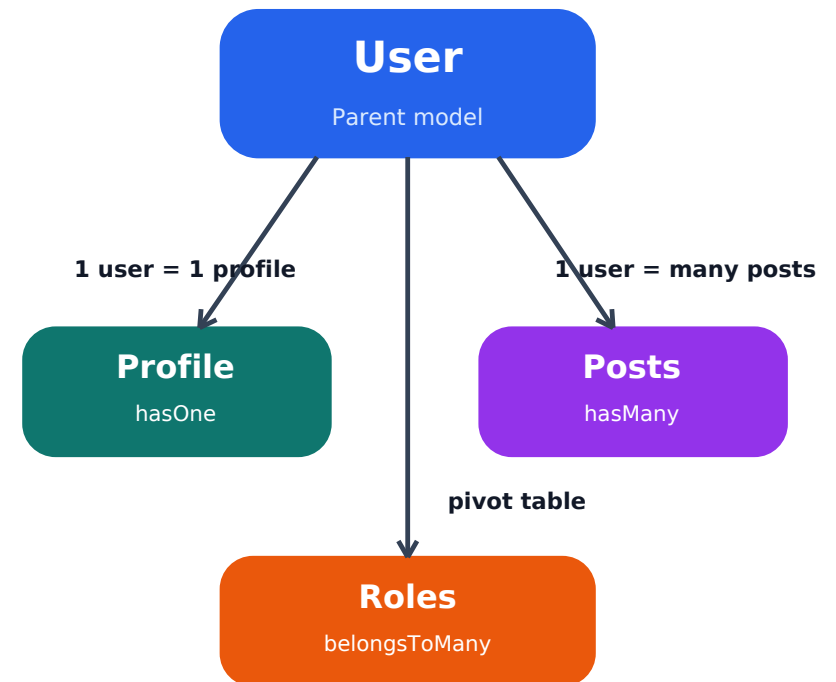
Ye PDF video ke description me dene layak hai. Isme video script wali lines nahi hain. Sirf clean diagrams, rules, model methods aur common mistakes hain - taki learners ko relationship visually clear ho.

Teaching formula

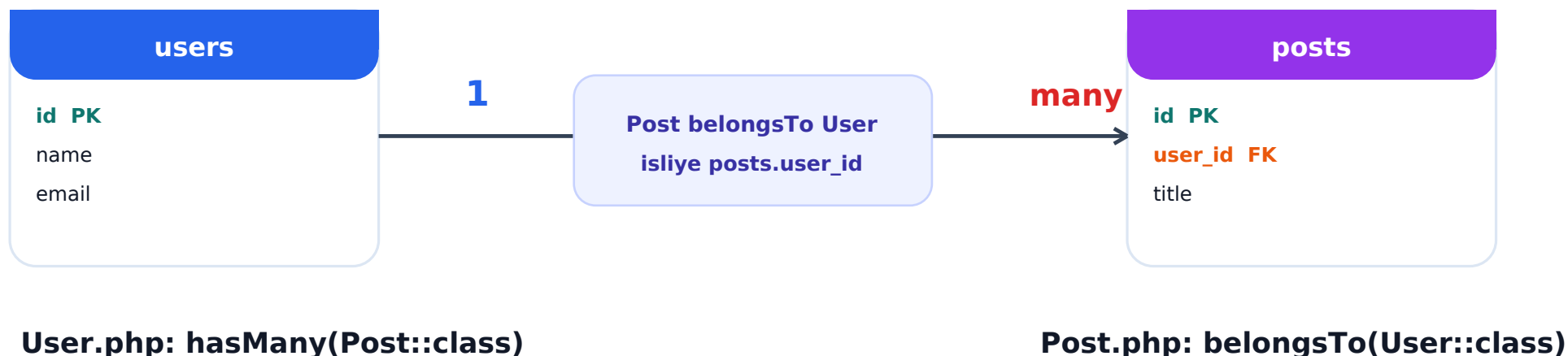
Har relationship ko 3 steps me samjhao: Database diagram, Model method, aur Actual access. Isse beginner ko join/foreign key ka confusion kam hota hai.

Best point

Relationship method ka kaam: database relation ko readable object style me access karna.



Core rule: belongsTo wali table me foreign key



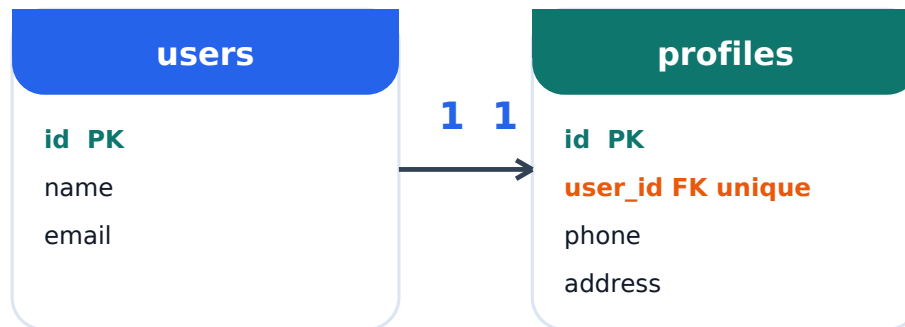
Simple Hindi

Jo table kisi doosri table ko point karti hai, us table me foreign key hoti hai.

Video me repeat karo

Foreign key direction clear ho gayi to relationships ka 70% concept clear ho gaya.

One to One: User has one Profile



1 user = 1 profile

Hindi explanation

One-to-one ka matlab ek record ka sirf ek related record. Example: ek user ki ek profile. Foreign key profiles table me hogi, kyunki profile user ko point karti hai.

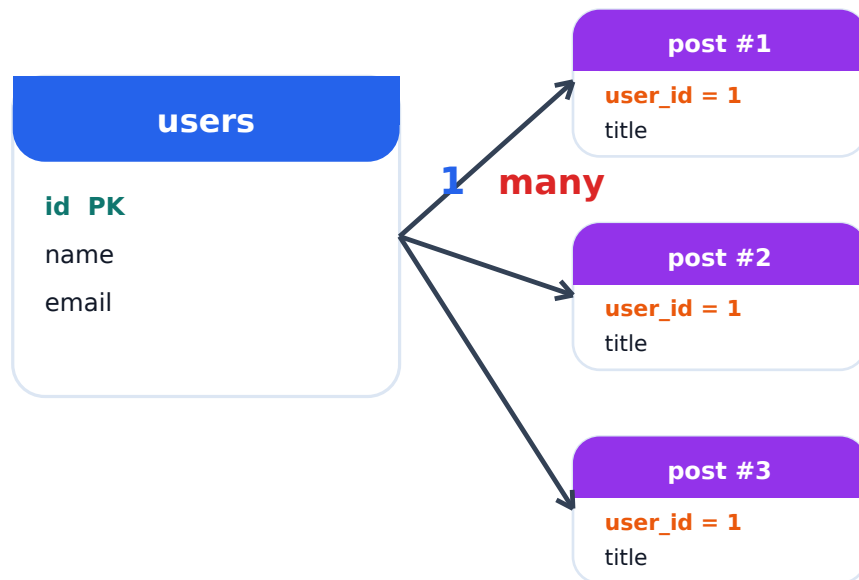
```
// User.php
public function profile() {
    return $this->hasOne(Profile::class);
}

// Profile.php
public function user() {
    return $this->belongsTo(User::class);
}
```

Access

User se profile: `$user->profile` | Profile se user: `$profile->user`

One to Many: User has many Posts



Hindi explanation

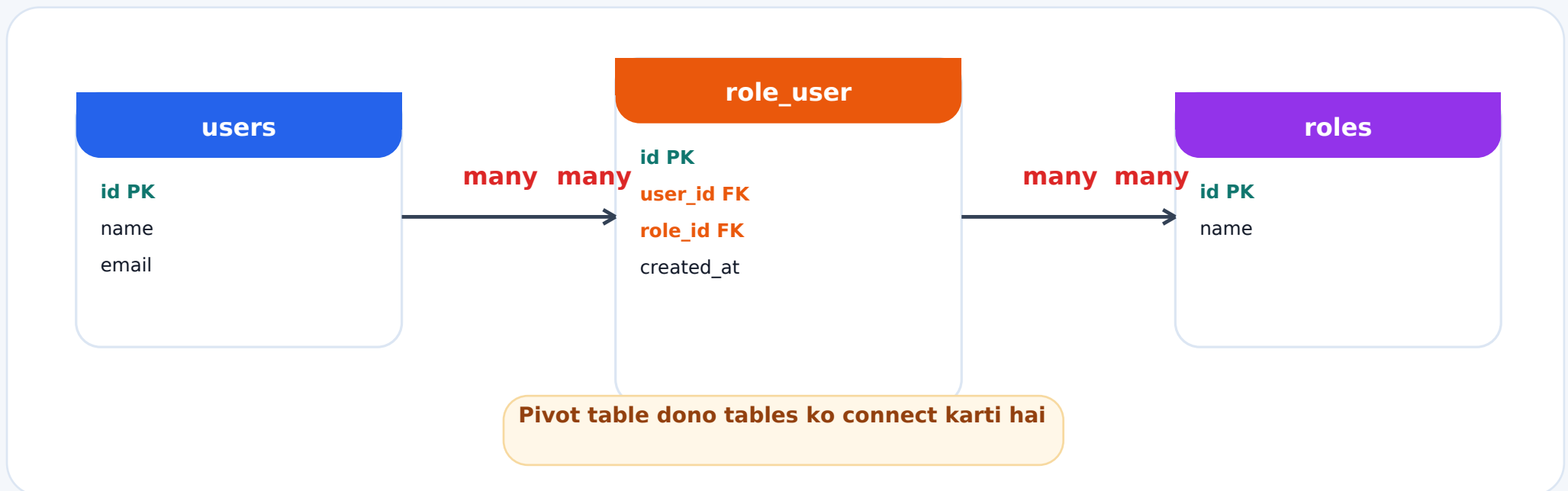
One-to-many me ek parent ke multiple child records ho sakte hain. Ek user ke many posts. Foreign key child table me hoti hai: posts.user_id.

```
// User.php
public function posts() {
    return $this->hasMany(Post::class);
}

// Post.php
public function user() {
    return $this->belongsTo(User::class);
}
```

Common mistake User model me belongsTo(Post::class) nahi. User parent hai, isliye hasMany(Post::class).

Many to Many: Users and Roles



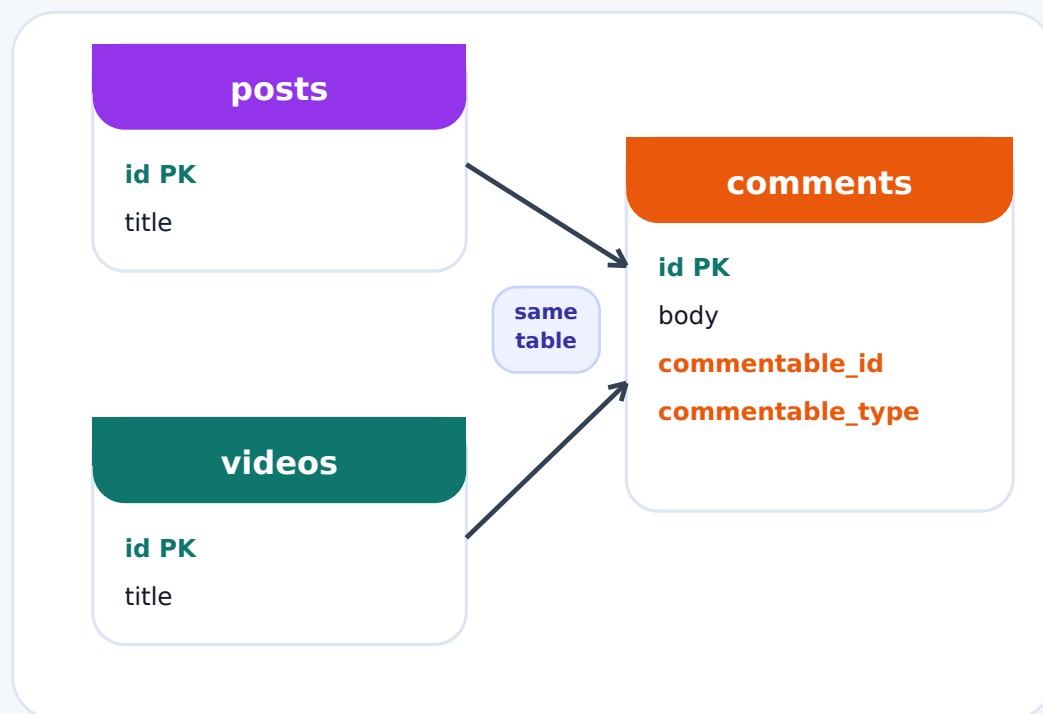
Model methods

```
User: belongsToMany(Role::class)
Role: belongsToMany(User::class)
```

Hindi explanation

Ek user ke multiple roles ho sakte hain, aur ek role multiple users ke paas ho sakta hai. Isliye beech me pivot table aati hai.

Polymorphic: same comments table



Hindi explanation

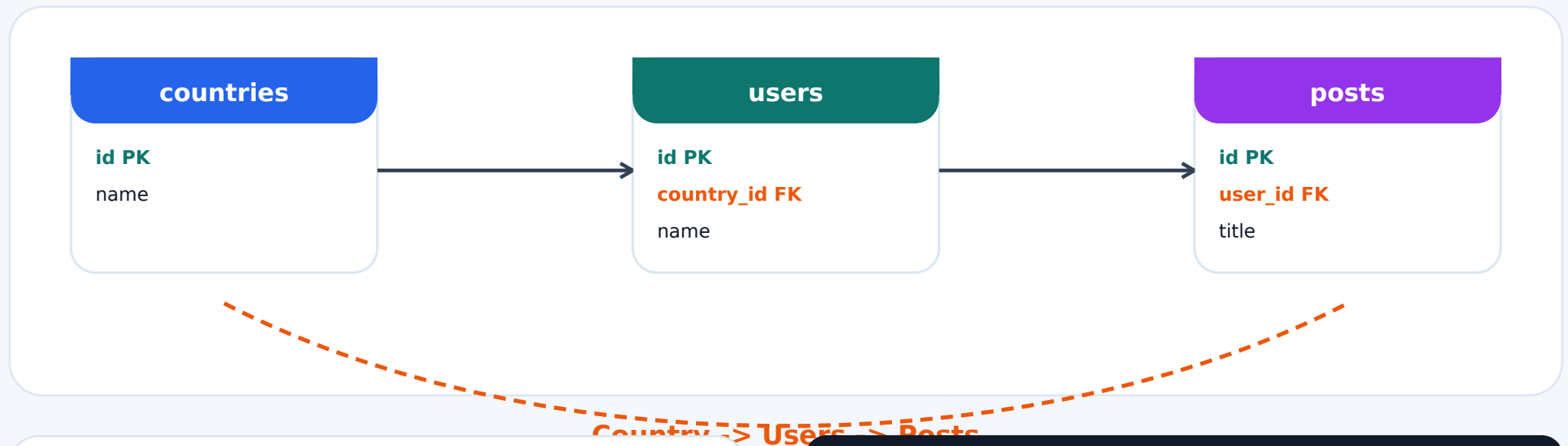
Polymorphic tab use hota hai jab ek hi table multiple models se connect ho. Comments post par bhi aa sakte hain aur video par bhi. Isliye comments table me commentable_id aur commentable_type hota hai.

```
// Post.php / Video.php
public function comments() {
    return $this->morphMany(Comment::class, 'commentable');
}

// Comment.php
public function commentable() {
    return $this->morphTo();
}
```

Teaching point commentable_type batata hai kis model ka comment hai. commentable_id batata hai us model record ki id.

Through: Country has many Posts through Users



Hindi explanation

Country directly posts se connected nahi. Beech me users table hai. Laravel shortcut: `hasManyThrough`.

```
// Country.php
public function posts() {
    return $this->hasManyThrough(Post::class, User::class);
}
```

N+1 query problem: Lazy vs Eager Loading

Wrong: Lazy loading

User::all()

1 query

user 1 posts

+1 query

user 2 posts

+1 query

```
$users = User::all();  
foreach ($users as $user) {  
    $user->posts;  
}
```

Right: Eager loading

User::with('posts')

query 1: users

query 2: posts

Fixed queries

Users kitne bhi ho, query control me

```
$users = User::with('posts')->get();
```

Explain like this Lazy loading me loop ke andar relationship call hota hai. Eager loading me Laravel related data pehle hi load kar leta hai.

Cheat sheet: kaunsa method kab?

Scenario	Parent side	Reverse side	Foreign key	Access
One to One	hasOne	belongsTo	Child table	<code>\${user}->profile</code>
One to Many	hasMany	belongsTo	Child table	<code>\${user}->posts</code>
Many to Many	belongsToMany	belongsToMany	Pivot table	<code>\${user}->roles</code>
Polymorphic	morphMany	morphTo	type + id columns	<code>\${post}->comments</code>

Mistake #1

Foreign key parent table me daal dena. Rule: belongsTo wali table me FK hoti hai.

Mistake #2

hasOne singular method: `profile()`.
hasMany plural method: `posts()`.

Mistake #3

Loop ke andar relationship call karke N+1 queries banana. Solution: `with()`.

Diagram based video flow

0:00

Hook

Foreign key kahan lagegi clear nahi, to relationships confusing lagenge. Aaj diagrams se clear karte hain.

2:00

One-to-one

User-profile diagram, model methods, phir \$user->profile.

6:30

Many-to-many

Pivot table center me rakho. Users aur Roles side me.

0:30

Core rule

belongsTo wali table me foreign key hoti hai. Users-posts diagram se prove karo.

4:00

One-to-many

User-posts diagram with multiple post boxes. Foreign key child table me.

9:00

N+1 problem

Lazy vs eager loading diagram. Ye advanced value deta hai.

Retention trick

Har 2-3 minute me ek common mistake dikhao. Mistakes audience ko hold karti hain, kyunki unhe lagta hai ab unki confusion solve hogi.